

Python Programming Guide

Python is a high-level language created by Guido van Rossum in 1991.
It emphasizes code readability with significant indentation.

Python is dynamically-typed and garbage-collected.
It supports multiple programming paradigms.

Key features: easy syntax, dynamic typing, and extensive libraries.
Used in web dev, data science, and AI.

Data Types

Python has several built-in data types.

Numeric: int, float, complex. Sequence: list, tuple, range.

Text: str. Mapping: dict. Set: set, frozenset.

Boolean: bool with values True and False.

Variables need no explicit declaration in Python.

Control Flow

Python supports standard control flow.

if/elif/else for conditional logic.

for loops for iterating over sequences.

while loops for repeated execution.

List comprehensions for concise list creation.

range function generates number sequences.

Functions

Functions are defined with `def` keyword.

Default arguments: `def power(base, exp=2)`

Lambda: `lambda x: x**2` for anonymous functions.

Functions are first-class objects in Python.

Modules are files with Python definitions.

Packages are collections of modules organized in directories.

File I/O

File operations use the with statement.

open for reading, writing, and appending files.

Exception handling with try/except/finally.

Common exceptions: TypeError, ValueError, KeyError.

The finally block always executes regardless of exceptions.

OOP

Python supports OOP with classes.

```
class Animal: def __init__(self, name)
```

Inheritance for code reuse between classes.

Encapsulation with `_protected` and `__private` attributes.

Polymorphism through duck typing.

Special methods like `__str__` and `__repr__`.

Data Processing

JSON: `json.dumps` and `json.loads` for data serialization.

Regex: `re.findall` and `re.match` for pattern matching.

Dates: `datetime` and `timedelta` for time manipulation.

Collections: `defaultdict`, `Counter`, `deque` for data structures.

Itertools: `chain`, `cycle`, `permutations` for iteration tools.